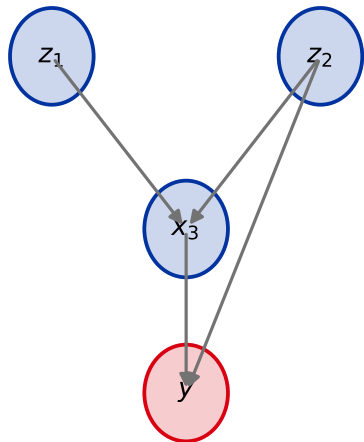


1. draw a random causal graph



2. draw random mechanisms

$$x_3 = \tanh(1.7z_1 - 0.9z_2) + \varepsilon$$

$$y = \mathbb{1}[0.8x_3 + 0.5z_2 + \varepsilon > 0]$$

the shapes, the weights and the noise are all sampled too

3. push noise through, read off a table

z_1	z_2	x_3	y
0.34	-0.66	0.97	0
-0.85	-0.52	-0.72	0
-0.21	-0.68	0.24	0
2.19	-0.38	1.16	1
-1.37	-1.37	-0.92	0
0.05	-0.90	0.50	0
0.20	-0.10	0.25	1
-1.80	-0.37	-0.90	0

This is ONE dataset. TabPFN's pre-training does this about 100 million times, with a different graph every time.